Response to Referee R2

We sincerely thank you for your time, effort, and valuable feedback. Your detailed comments and constructive suggestions have significantly improved the paper. We have carefully considered each point and prepared a thorough, point-by-point response. Below, we respond to each comment individually; for ease of reading, the original comments are italicized and displayed in red. Before addressing each comment, we begin with a brief summary of the major revisions.

- **Extended sample.** Following R1's and R6's suggestions, we extended our panel data coverage from seven years (1930–1936) to fifteen years (1926–1940). This extension serves two primary purposes:

  1. Examining additional pre-treatment years (1926–1932 rather than just 1930–1932) enables a more rigorous evaluation of the parallel trends assumption underlying our difference-in-differences design.
  2. The recession of 1937–1938 provides a natural placebo test: if our gold-clause exposure measure captured something other than the mechanism we propose—such as general sensitivity to economic downturns—we would expect it to affect investment during this later period of distress as well.

  Figure R.I (Figure 3 in the revised manuscript) plots the year-by-year interaction coefficients from our main investment regression over the extended sample. The figure confirms the absence of pre-trends before 1933, sharp negative effects during the 1933–1934 litigation period, and no effect during the 1937–1938 recession.

- **Data reconstruction.** While extending the sample, we identified that some scans purchased from Mergent were of poor quality, leading to firm omissions and data entry errors. To address these concerns, we undertook a complete data reconstruction:

  1. We scanned, in high resolution, the Moody's Industrial Manuals for all sixteen years, ensuring complete legibility of all pages.

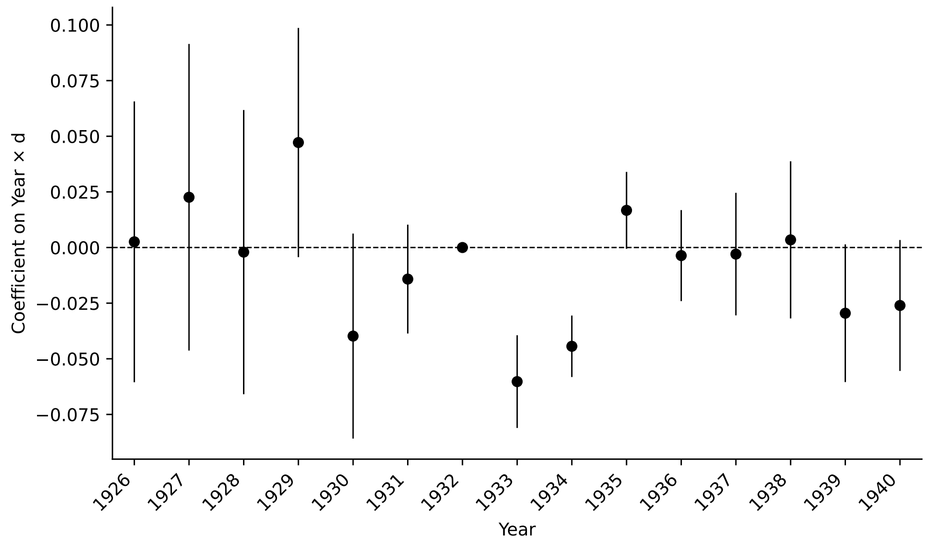

Figure R.I. Gold clause effect on investment by year

Notes. This figure plots point estimates and 95% confidence bands from a regression of net investment on Q and $\tilde{d}$ interacted with year indicators, including firm and year fixed effects.

2. We recollected the entire dataset using a double-blind procedure in which two independent research assistants collected data separately, with discrepancies resolved by a third party.

3. We implemented extensive quality control checks to identify and correct any remaining issues.

This reconstruction process required nearly two years to complete but was essential for ensuring the reliability of our findings.

- **Gold clause exposure measure.** Following R1's suggestion, we replaced total debt with long-term liabilities in the denominator of our gold exposure measure, $d_{j,t}$, now defined as the ratio of gold-clause bonds to long-term liabilities (corporate bonds, preferred shares, and bank loans) for firm j at time t:

$$d_{j,t} = \frac{\text{Gold-denominated long-term liabilities of firm } j \text{ in year } t}{\text{Long-term liabilities of firm } j \text{ in year } t}.$$

- **Addressing endogeneity concerns.** Following R6's suggestion, we fix each firm's gold exposure at its 1930 value for all subsequent years to address potential endogeneity from strategic adjustments following Britain's departure from the gold standard in September 1931. Formally:

$$\tilde{d}_{j,t} = \begin{cases} d_{j,t}, & \text{if } t \leq 1930 \\ d_{j,1930}, & \text{if } t > 1930 \end{cases}.$$

This measure, $\tilde{d}$, is our main explanatory variable throughout the revised analysis.

- **Empirical design.** Following R1's, R2's, and R6's suggestions, we now use the full 1926–1940 panel and estimate the effects of gold clause exposure and its reversal in a single regression.

- **Debt overhang framing.** Following R1's and R2's suggestions, we have substantially revised the paper's conceptual framing. We no longer use the term "leverage risk" in the paper title or text and instead interpret the events of 1933–1934 through the lens of dynamic debt overhang theory (Hennessy, 2004). This framing more precisely captures the economic mechanism linking gold clause exposure to investment decisions.

- **Firm-level uncertainty literature.** Following R2's suggestion, we have repositioned the paper within the firm-level uncertainty literature (Bloom, 2009; Baker et al., 2016; Hassan et al., 2019). The uncertainty surrounding the gold clause represents a policy shock of the type studied extensively in this growing literature. This reframing highlights that our gold clause exposure measure, $\tilde{d}_{j,t}$, captures firm-level variation in exposure to policy uncertainty—derived from balance sheet data rather than earnings call transcripts as in (Hassan et al., 2019). We are grateful to R2 for this suggestion, as it clarifies how our historical setting illustrates the real effects of policy uncertainty during a pivotal episode.

1. Framing:

The paper is closely related to firm-level uncertainty after Brexit studied by Hassan et al (2019, not cited). While a whole cottage industry has followed the firm-level approach pioneered by Hassan et al, I am not aware of a paper that focuses on an uncertainty shock to financial arrangements. Therefore, identifying a case of firm-level uncertainty about their financial contracts is a significant contribution to the literature. While the authors briefly point to Bloom (2009), the authors should highlight these firm-level differences in uncertainty much more and emphasize that these differences matter for aggregate outcomes during the Great Depression. Given the nature of the shock and the resulting litigation, I don't think the effects are due to 'leverage risk' but to uncertainty about leverage. By reframing the paper and linking it to the literature on firm-level uncertainty, the paper would have a larger impact.

Thank you for this insightful comment, which has substantially shaped our revised manuscript. Following your suggestion, we have engaged more deeply with the literature on policy uncertainty and its effects on firm outcomes (Bloom, 2009; Baker et al., 2016; Hassan et al., 2019). In particular, we read Hassan, Hollander, van Lent, and Tahoun (2019) carefully, along with the broader literature that has followed their work. This engagement has clarified how our contribution fits within this active area of research.

Hassan, Hollander, van Lent, and Tahoun (2019) pioneers a firm-level approach to measuring political uncertainty, using textual analysis of earnings calls to identify heterogeneous exposure to policy shocks across firms. A key insight from this work is that aggregate measures of uncertainty can mask substantial firm-level variation in exposure, and that this variation has real consequences for firm decisions. We now recognize that our setting offers a complementary approach to measuring firm-level exposure: rather than inferring exposure from managerial discussions, we measure it through the gold content of firms' liabilities. This parallel is now explicit in the literature review, where we write that "in the same spirit as Hassan, Hollander, van Lent, and Tahoun (2019), who measure firm-level exposure to policy shocks using textual analysis of earnings call transcripts, we identify firm-level exposure to legal uncertainty using the gold content of firms' liabilities."

More broadly, the influence of your suggestion can be seen throughout the paper, starting with its new title: "Debt Overhang During Policy Uncertainty: The Case of Gold Clauses in the 1930s." In the introduction, we now position our study as "closest to the literature on firm-level political uncertainty pioneered by Hassan, Hollander, van Lent, and Tahoun (2019)" and emphasize that the abrogation represents "political intervention in private debt contracts." While Hassan, Hollander, van Lent, and Tahoun (2019) documents the real effects of uncertainty in modern episodes such as Brexit, U.S. elections, and Italian constitutional reforms, our study examines how similar mechanisms shaped firm decisions during the Great Depression—a pivotal historical episode that complements the modern evidence.

As you note in the second paragraph of your report, by studying the effects of debt overhang, we "follow a well-trodden path." However, the unique features of our empirical setting also enable us to study how policy uncertainty that *directly* impacts the right-hand side of the firm's balance sheet (through leverage uncertainty) also *indirectly* impacts the left-hand side (through firms' investment decisions). While we interpret our results through the lens of debt overhang, in this episode it takes a unique form shaped by political decisions affecting firm-level choices. We now state in the introduction that "we contribute to this literature by showing how an uncertainty shock to future liabilities delayed the recovery of investment during the Great Depression."

2. There are several issues with the empirical specification:

a. Why don't the authors use all data in the regression and estimate the treatment and the reversal in one regression on a balanced panel? This way the firm effects are consistently estimated.

We agree and have substantially revised our empirical approach following your suggestion. In the revised manuscript, we now estimate the treatment and reversal effects jointly in a single regression spanning the entire 1926–1940 sample period. For convenience, we report estimation results of the baseline investment regression depicted in Figure R.I (Figure 3 in the revised manuscript) at the beginning of our response. This unified specification allows firm fixed effects to be consistently estimated using all available observations, addressing your concern about inconsistent estimation from separate regressions.

Specifically, our baseline specification in Table 3 (Section 5.2) interacts gold clause exposure $\tilde{d}$ with year indicators for each year from 1926 to 1940, using 1932 as the omitted category. This design presents year-by-year effects rather than grouping into two-year intervals, providing a more transparent view of the treatment dynamics. The coefficient pattern now shows: (i) no pre-trends from 1926–1931, (ii) significant negative effects in 1933 and 1934, and (iii) reversal after the 1935 Supreme Court decision. Following R6's suggestion, to strengthen identification we fix $\tilde{d}$ at its 1930 value—before Britain's departure from the gold standard—for all subsequent years, as detailed in Section 4.

Regarding balanced panels, we conduct robustness checks in Internet Appendix Table IA.12. We repeat our baseline regression using: (i) firms with uninterrupted data for 1930–1936 (Column 2, $N = 6,064$), (ii) 1929–1940 (Column 3, $N = 5,435$), and (iii) the full 1926–1940 period (Column 4, $N = 3,630$). While requiring a fully balanced panel for 15 years substantially reduces the sample, the interactions of $\tilde{d}$ with the 1933 and 1934 indicators remain negative and statistically significant in most specifications. For instance, in the 1930–1936 balanced panel, the coefficients are -0.035 and -0.033 for 1933 and 1934, respectively, both significant at the 1% level.

b. *There seems to be entry and exit, so it would be useful to know whether firms with gold clauses were more likely to exit. The answer is likely to be no given the results for profits. This would help the authors to the extent that it complements the profit non-result, as one would expect from an uncertainty shock. More generally, the authors should consider the effects of entry and exit in their calculations in table 11.*

Consistent with your conjecture, we examined the data and found no evidence that gold-exposed firms were more likely to exit. As you note, this is consistent with the profit results in Table 4 (Column 4), which show no differential profitability for gold-exposed firms in 1933–1934. The balanced panel results in Internet Appendix Table IA.12 provide further indirect evidence that differential exit is not driving our findings, as we find similar results when restricting to firms with continuous data.

Regarding the aggregation exercise (Table 7 in the revised manuscript, Section 5.6), we compute total investment using only firms with data for two consecutive years. Specifically, the formula for the gold clause effect weights each firm's contribution by its fixed capital in the previous year and requires presence in both years t and $t-1$. This design ensures that our aggregate estimates reflect changes in investment behavior among continuing firms rather than compositional shifts from entry or exit.

c. *If available, I would like to understand whether the presence of the gold clause is related to the bond origination date. Specifically, I would expect gold clauses to be more likely to be present, the later the bond was issued (closer to 1933) since some countries had already left the gold standard. If true, the authors should also provide evidence that the timing of issuances does not affect their results (selection).*

A key fact to keep in mind: in 1932, about 97% (99%) of the corporate bonds in our sample by count (by outstanding par value) had the gold clause. In our original submission, we had this information in a footnote, but now feature it more prominently in the paper.

We nonetheless examined the data looking for meaningful time-series variation in the presence of the gold clause as a function of the issuance year. The data reveals that the gold clause was ubiquitous across all issuance cohorts. Given this near-universal prevalence, variation in issuance timing should not meaningfully affect our estimates of gold clause exposure effects.

We now emphasize the near-universal prevalence of gold clauses throughout the paper. In the introduction, we state: "around 97% of outstanding corporate bonds were subject to a gold clause." Section 3.1 discusses the emergence and adoption of the gold clause, noting that "the clause is present in the vast majority of corporate bonds (97% in our sample)." We also include President Hoover's 1932 statement that "most of our industrial and all of our Government, most of our State and municipal bonds, and most other long-term obligations are written as payable in gold," which reflects the contemporary understanding that gold clauses were standard practice. Section 4 reiterates this point when defining gold clause exposure in equation (1). Table 2 reports detailed statistics on the number of bonds and the fraction containing gold clauses.

d. *In table 6 the authors control for several balance sheet variables, some of which could be changing due to the gold clauses. The authors should present results that show whether gold clauses affected leverage and cash holdings as one response to uncertainty would be increasing cash and reducing leverage. This could be shown in table 6 and columns 2-8 could be relegated to the appendix.*

Thank you for this suggestion. We now include results for profits, cash, and leverage in Table 4 of the revised manuscript, discussed in Section 5.3. We find no evidence that high-$\tilde{d}$ firms increased cash holdings or reduced leverage in 1933–1934 relative to other firms.

We agree that the precautionary motive you describe is intuitive: firms facing uncertainty might build cash buffers or reduce leverage. However, we also believe this motive competes with the incentive under debt overhang to extract resources from the firm. When equityholders anticipate losing their claim with higher probability, they have limited incentive to strengthen the balance sheet and may instead prefer to pay out value. The evidence is consistent with the view that the two effects roughly cancel out.

e. *Following up on the previous point, the authors could present more results on financing constraints. In addition to credit rating in table 5, showing whether the effect is larger for smaller firms, firms with low cash holdings, firms with higher book leverage would be standard in a paper like this.*

We agree and now present additional results with triple interactions of gold clause exposure d, year indicators, and indicators for small firms, low cash firms, and high-leverage firms (Internet Appendix Table IA.13). The effects are larger for smaller firms, likely reflecting their more limited funding alternatives. While we do not find significant effects for cash or leverage, we do find that firms with lower credit ratings were more affected (Table 5), which is perhaps a more direct measure of how uncertainty differentially impacted firms through debt overhang.

f. *I would also like to see some additional specification for the intensive margin: consider a dummy for high exposure at different cutoffs, d>0.2, d>0.4.*

We have conducted an additional test regressing investment on dummies for positive $\tilde{d}$, above-median $\tilde{d}$, and above-75th percentile $\tilde{d}$, with results reported in Internet Appendix Table IA.16. In this design, we interact these $\tilde{d}$ indicators with four time dummies: 1926–1932 (Before), 1933, 1934, and 1935–1940 (After), using Before as the omitted category. Column 1 shows that the results hold at the extensive margin: firms with $\tilde{d} > 0$ invested significantly less than $\tilde{d} = 0$ firms in 1933 and 1934 relative to the pre-period. We agree that non-linear effects are plausible, and consistent with this, we observe negative point estimates for high-$\tilde{d}$ interactions, though the coefficients are imprecisely estimated. For example, above-median $\tilde{d}$ firms show an additional negative effect in 1933 (Column 2), while the negative estimate for 1934 is not statistically significant. We briefly refer to these results in a footnote at the end of Section 5.4.

g. *The authors need to cluster standard errors on a time dimension. Given the number of years in the sample, clustering on the industry-year level is appropriate.*

We agree that accounting for time-series correlation in the errors is important, and we now two-way cluster standard errors by firm and year in all specifications. We chose firm-year clustering rather than industry-year clustering for the following reasons.

First, firm-level clustering addresses serial correlation within firms over time, which is a first-order concern in panel settings with firm fixed effects. Second, clustering by year accounts for cross-sectional correlation driven by common shocks—such as the gold clause resolution itself—that affect all firms simultaneously. With 15 years in our extended sample, we have sufficient time-series variation for year-level clustering to be well-behaved.

h. In table 11, I would also show a panel for firms with d=0.

In the aggregation exercise (Table 7 in the revised paper), we now report the aggregated investment rate for $d = 0$ firms in Panel C. As expected, their net investment rates are less negative compared to $d > 0$ firms.

3. When reframing the paper, please shorten the introduction. The current version is too long and meandering.

When reading the draft again with some distance, we fully agreed with you. In particular, the part where we used to lay out the modern day relevance of our study felt out of place and tangential. As a result, we removed this section, which cut the length of the introduction by almost three pages. We also reduced the length of the introduction by adopting a less detailed description of the robustness checks we perform in the paper. We hope you will agree with us that, following these modifications, the introduction is less meandering and reads better overall.

Thank you again for your constructive and insightful comments! The new analyses, guided by your suggestions, have substantially strengthened the empirical evidence and improved both our understanding of the mechanism and the positioning of the paper within the literature.

REFERENCES

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